

XG-NID: Dual-Modality Network Intrusion Detection using a Heterogeneous Graph Neural Network and Large Language Model

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ABSTRACT

In the rapidly evolving field of cybersecurity, the integration of flow-level and packet-level information for real-time intrusion detection remains a largely untapped area of research. This paper introduces "XG-NID," a novel framework that, to the best of our knowledge, is the first to fuse flow-level and packet-level data within a heterogeneous graph structure, offering a comprehensive analysis of network traffic. Leveraging a heterogeneous graph neural network (GNN) with graph-level classification, XG-NID uniquely enables real-time inference while effectively capturing the intricate relationships between flow and packet payload data. Unlike traditional GNN-based methodologies that predominantly analyze historical data, XG-NID is designed to accommodate the heterogeneous nature of network traffic, providing a robust and real-time defense mechanism. Our framework extends beyond mere classification; it integrates Large Language Models (LLMs) to generate detailed, human-readable explanations and suggest potential remedial actions, ensuring that the insights produced are both actionable and comprehensible. Additionally, we introduce a new set of flow features based on temporal information, further enhancing the contextual and explainable inferences provided by our model. To facilitate practical application and accessibility, we developed "GNN4ID," an open-source tool that enables the extraction and transformation of raw network traffic into the proposed heterogeneous graph structure, seamlessly integrating flow and packet-level data. Our comprehensive quantitative comparative analysis demonstrates that XG-NID achieves an F1 score of 97% in multi-class classification, outperforming existing baseline and state-of-the-art methods. This sets a new standard in Network Intrusion Detection Systems (NIDS) by combining innovative data fusion with enhanced interpretability and real-time capabilities.

1. Introduction

In the dynamic and increasingly complex landscape of cybersecurity, Network Intrusion Detection Systems (NIDS) play a crucial role in protecting systems and networks from a wide range of cyber threats (Mallick and Nath, 2024). The growing significance of cybersecurity is underscored by the rise in cyber-attacks driven by rapid digital expansion (Mwangi, 2024). In an era marked by unprecedented interconnectedness and technological advancement, the cybersecurity landscape faces an escalating challenge: the relentless wave of cyber threats targeting critical infrastructures, sensitive information, and the core functions of society (Farrukh et al., 2024a). As the digital landscape expands, so does the complexity and sophistication of cyber-attacks, necessitating a paradigm shift in defensive strategies (Cunningham, 2020). According to Gartner, by 2025, 30% of businesses in critical infrastructure will face a security breach, potentially leading to the shutdown of mission-critical cyber-physical systems (Gartner, Inc., 2021), highlighting the critical need for improved defensive measures.

Traditional machine learning-based NIDS methodologies can be broadly categorized into two types: those that analyze flow information and those that scrutinize packet-level information (Farrukh et al., 2023). Flow-based NIDS analyze aggregated data about network traffic, such as the volume of data transferred between endpoints, the duration of connections, and the frequency of interactions. This approach is efficient for identifying patterns and anomalies at a higher level, such as Distributed Denial of Service (DDoS) attacks, which are characterized by unusually high volumes of traffic (Specht and Lee, 2004). However, flow-based systems can miss more nuanced attacks that are embedded within the payload of individual packets (Umer et al., 2017).

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Conversely, packet-based NIDS focus on the contents of individual packets, examining payloads for signatures of known exploits, malware, and other malicious content. This method excels at detecting attacks that rely on specific payload characteristics, such as SQL injection or buffer overflow attacks (Farrukh et al., 2022). Despite their detailed inspection capabilities, packet-based systems can be overwhelmed by high volumes of traffic and may fail to identify broader traffic patterns indicative of certain attacks.

The inherent limitations of relying solely on either flow or packet-level information highlight the need for an integrated approach. Certain sophisticated attacks can exploit these limitations to bypass detection. For instance, an SQL injection attack, which embeds malicious SQL commands within a packet's payload, might evade a flow-based NIDS because it does not generate significant anomalies in traffic patterns (Umer et al., 2017). Conversely, flow-based attacks can slip through packet-level NIDS because they do not involve malicious payloads but rather manipulate the volume and frequency of packets (Tan et al., 2014). This dichotomy underscores a critical gap in current machine learning-based NIDS technologies. By failing to leverage the complementary strengths of both flow and packet-level information, existing systems leave networks vulnerable to a range of attack vectors. Addressing this gap requires a novel approach that combines these two types of information into a unified framework, enabling more comprehensive threat detection.

In response to these challenges, our paper proposes a novel method of multi-modal data fusion that integrates both flow and packet-level information, represented in a unique heterogeneous graph format. This approach leverages Graph Neural Networks (GNNs) to process and analyze the heterogeneous data, capturing the intricate relationships and patterns within network traffic. By representing network traffic as graphs, where nodes and edges encapsulate both flow and packet attributes, we can harness the strengths of both modalities of data. This fusion allows our system to detect a broader spectrum of cyber-attacks, improving overall detection accuracy. The graph-based representation also facilitates the learning of complex network patterns, which are essential for identifying sophisticated threats.

Furthermore, we emphasize the importance of contextual explainability in cybersecurity. Traditional NIDS outputs are often cryptic, making it difficult for cyber analysts to understand and respond to threats effectively (Abou El Houda et al., 2022). By integrating explainability using contextual information from large language models (LLMs), our approach provides clear, actionable insights into detected threats, detailing the nature of the attack, affected components, and recommended mitigation strategies.

In short, this paper emphasizes the importance of data fusion in network security and proposes a novel framework that integrates flow and packet-level information using GNNs, along with contextual and explainable inferences. By bridging the gap between these two data modalities and leveraging their combined information, our approach offers a robust and comprehensive solution to network intrusion detection. This framework not only enhances the detection of threats but also provides actionable insights that can lead to the development of intelligent systems capable of making autonomous decisions on appropriate responses, thereby paving the way for improved cybersecurity measures. The main contributions of this work are as follows:

- 1) We present a novel framework "XG-NID" that fuses flow-level and packet-level data modalities in network traffic, providing contextual, explainable, and actionable insights to enhance model interpretability and support informed decision-making.
- 2) We introduce an innovative method for fusing dual modalities of network traffic—flow and packet-level information—using heterogeneous graph structures for comprehensive analysis.
- 3) We develop a heterogeneous graph neural network (HGNN) based on graph-level classification for NIDS, enabling real-time inference by utilizing dual-modality data and incorporating the heterogeneous nature of network traffic.
- 4) We develop "GNN4ID," (Farrukh et al., 2024b) an open-source tool for extracting and transforming raw network traffic into our proposed heterogeneous graph structure, integrating flow and packet-level data.
- 5) We propose a new set of flow features based on temporal information to aid in producing better contextual and explainable inferences.
- 6) We present a methodology for providing detailed and human-readable explanations of the proposed HGNN model results by combining contextual information from LLMs.
- 7) We provide a quantitative comparative analysis of our proposed methodology with other baseline and comparable approaches to highlight its effectiveness.

The remainder of this paper is structured as follows: Section 2 reviews related works, covering the application of GNNs in NIDS and the role of explainable artificial intelligence (XAI) in NIDS. Section 3 details the adopted

methodology, including the dataset, proposed framework, its integral components, and their operation. Section 4 presents the results and provides a comparative analysis. Section 5 concludes the paper and outlines potential future work.

2. Related Works

In this section, we review the existing literature in two parts: the application of GNNs in NIDS and the role of XAI in the NIDS domain. Notably, there is a significant gap in the research concerning the integration of packet-level and flow-level information for both inference and explainability. Our work aims to address this gap by leveraging data fusion through heterogeneous graph structures, extracting contextual information from both packet and flow levels, and performing **graph-level classification** to enable real-time inference with explainability. This comprehensive approach not only enhances the detection capabilities but also provides actionable insights, advancing the field of network security.

2.1. Graph Neural Network

The majority of the existing literature on **GNN-based NIDS focuses on node-level and edge-level classification**. These studies are primarily aimed at analyzing historical network traffic to understand past attacks or to visualize network activity in a more user-friendly manner, often lacking real-time inference capabilities (Messai and Seba, 2023). Further, most of these works employ homogeneous graph representations of network traffic, which do not fully exploit the heterogeneous nature of network data. This leaves a gap in leveraging the heterogeneity of network traffic and providing real-time inference. We discuss several notable works herein, highlighting their contributions and limitations.

Lo et al. (2022) presented "**E-GraphSAGE**" to **classify network flows using edge-level classification**. The model captures edge and topological information in IoT networks for classification. The authors enhanced the GraphSAGE (Hamilton et al., 2017) algorithm to **directly exploit the structural information of the network flow and encode it in a graph**. Despite these advancements, the scalability and real-time inference capabilities remain questionable due to the **computational complexity of analyzing complete flow information**. Chang and Branco (2024) further enhanced **E-GraphSAGE by integrating residual learning and an attention mechanism to increase efficiency**. They utilized E-GraphSAGE with residual learning to target minority class imbalance and introduced an edge-based residual graph attention network (E-ResGAT) to improve efficiency. However, the reliance on fixed neighborhood edge sampling and attention mechanisms might still **face challenges in highly dynamic network environments**.

Zhou et al. (2020) proposed a supervised approach based on a **Graph Convolutional Network (GCN) and network traces**. They considered only the topological structure of the graph, omitting edge features and initializing node features as a vector of ones. **This method can efficiently handle large graphs but may miss critical edge information that could enhance detection accuracy**. Similarly, Zhang et al. (2022) suggested using a GCN for **botnet detection**. They used 12 GCN layers to capture long-term dependencies in large botnet architectures. **However, very deep GCN models are prone to over-smoothing, which can diminish the model's ability to distinguish between classes**.

Altaf et al. (2023) presented a comprehensive **GNN-based NIDS model** capable of **capturing relations in the network graph and combining both node and edge features** to identify abnormal traffic behavior. **This approach uses IP addresses and port numbers to represent Internet of Things (IoT) sessions as nodes and network flows as edges, which helps mitigate multiple attack vectors at the application and network layers. However, attacks dependent on payload content can still evade detection**. A more general approach to detecting intrusions with a heterogeneous graph is proposed in (Pujol-Perich et al., 2022). The graph is built based on network flows, creating separate nodes for the source host, destination host, and the flow itself. This structure, combined with a **nonstandard message-passing neural network (MPNN)**, improves the model's ability to learn embeddings from flows. Despite its promising accuracy, the model was tested on a heavily pre-processed dataset, raising questions about its performance on more balanced and varied data.

Cao et al. (2021) proposed representing packet traffic using a spatio-temporal graph to model features that vary with time. Their method aims to detect DDoS attacks in software-defined networking (SDN) environments using a Spatio-Temporal Graph Convolutional Network (ST-GCN) (Yu et al., 2017). While this approach effectively captures temporal dynamics, its applicability to a wider range of attacks remains to be seen. **Premkumar et al. (2023) is the only paper we found that effectively integrates both flow and packet information utilizing Graph Representation Learning (GRL). Their approach first generates packet-level embeddings from graph representations, then combines these with flow-level**

features for final prediction. While this method captures detailed packet information and broader flow characteristics, it lacks direct fusion of these features, potentially leading to increased computational overhead due to the two-step processing required.

In short, while existing GNN-based NIDS models offer various methods for network intrusion detection, they often focus on either node or edge-level classification and typically lack the integration of heterogeneous information sources. Our proposed approach addresses these shortcomings by leveraging both packet and flow-level data within a heterogeneous graph framework, enabling more comprehensive and real-time intrusion detection. By utilizing graph-level classification, our model can better capture the complex relationships inherent in network traffic and provide robust defense mechanisms against a broader range of cyber threats.

2.2. Explainable Artificial Intelligence

Recent research in XAI has been actively applied to cybersecurity, particularly in specialized use cases like network intrusion detection and malware identification. Han et al. (2021) developed the DeepAID framework to interpret unsupervised deep learning-based anomaly detection systems for cybersecurity. This approach helps security analysts understand why a certain sample is considered anomalous by comparing the anomaly to a normal reference data point.

XAI holds promise in enhancing the adoption of machine learning (ML) models within existing cybersecurity frameworks. However, several challenges still need to be addressed (Nadeem et al., 2023), particularly in making explanations more understandable for users and reducing the opacity of ML models within NIDS frameworks. To address these issues, Alwahedi et al. (2024) explored the integration of LLMs with XAI techniques. Consequently, recent research papers have focused on adding an LLM layer to the existing frameworks.

Ziems et al. (2023) introduced LLM-DTE (Large Language Model Decision Tree Explanations), a method that utilizes LLM to generate natural language explanations for decision tree-based NIDS. Khediri et al. (2024) proposed an approach that integrates SHAP (SHapley Additive exPlanations) values with LLMs to generate human-understandable explanations for detected anomalies. These SHAP values provide a detailed explanation of individual predictions by measuring the impact of each feature on the final outcome (Moustafa et al., 2023). Their methodology, applied to the CICIDS2017 dataset, demonstrated how this combination of SHAP values and LLM can offer coherent responses regarding influential predictors of model outcomes. Similarly, Ali and Kostakos (2023) developed HuntGPT, a specialized intrusion detection dashboard that incorporates XAI frameworks like SHAP with a Random Forest classifier trained on the KDD99 dataset to enhance model interpretability.

Despite these advancements, significant issues remain regarding the scalability, efficiency, and human-centric aspects of XAI-driven cybersecurity solutions. Current explanations based on SHAP values often lack the necessary contextual and temporal information, which is crucial for accurately identifying and explaining time-based attacks. Time-based attacks, such as DDoS attacks, unfold over specific periods, where the timing and sequence of network events are quite important. A more effective explanation would incorporate temporal information, with features highlighting a rapid influx of requests to the targeted server from specific IPs within a brief timeframe, providing a clearer understanding of why the attack was detected.

Explaining payload-specific attacks based on network flow remains a challenge. The actual signature of payload-specific attacks, such as SQL injection, lies within the payload itself. These attacks can demonstrate benign behavior in network flow, with their malicious intent residing in the packet content. Thus, explanations must consider the payload to clearly convey why a prediction was made. Current methods that explain such attacks solely based on flow data fail to capture the critical details within the payload, leading to incomplete or inaccurate explanations. These shortcomings are addressed in this paper by enhancing the feature extraction process to include both contextual and temporal information for time-based attacks, and by ensuring payload-specific attacks are explained based on their content.

3. Methodology

In this section, we present the methodology employed to develop and evaluate our proposed framework for network intrusion detection. The framework leverages the fusion of flow and packet-level information through a HGNN to provide real-time, explainable, and actionable insights. We begin by providing details of the proposed framework, including the feature extraction and development, graph construction, and the integration of explainability through contextual inference. This is followed by details of the dataset used for training and testing. The methodology outlines the key components that contribute to the effectiveness of our approach, ensuring comprehensive analysis and robust intrusion detection capabilities.

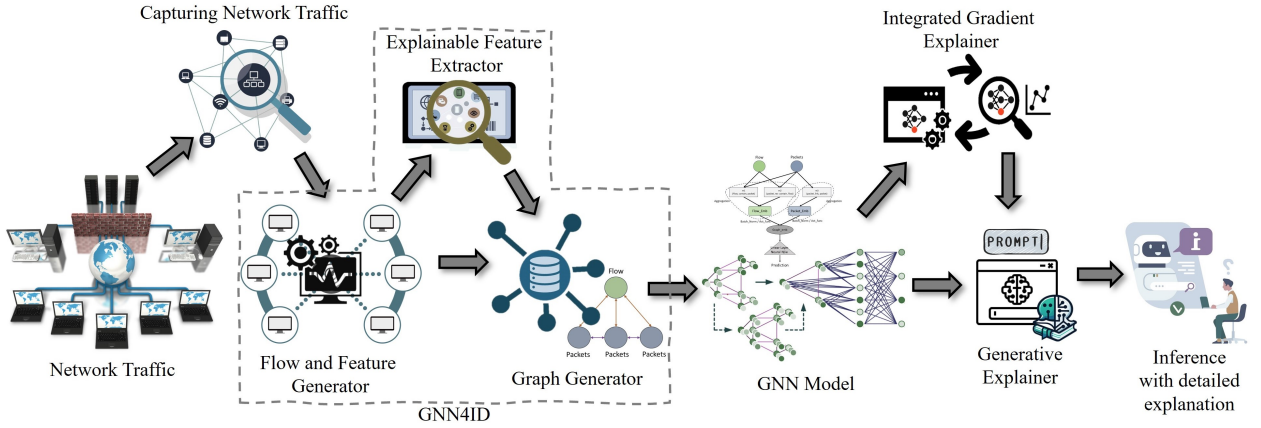


Fig. 1. Illustration of the proposed framework "XG-NID". The framework, depicted here, demonstrates its capability to better identify network attacks either based on flow or packet-level information with detailed contextual information and potential remedial actions. The framework comprises of six key components: Flow and Feature generator, Explainable Feature Extractor, Graph Generator, GNN Model, Integrated Gradient Explainer, and Generative Explainer.

3.1. Proposed Framework

The proposed framework "XG-NID" presents a novel approach to network intrusion detection by integrating flow and packet-level information into a heterogeneous graph structure, enabling a more comprehensive analysis of network behavior. This fusion of multiple data modalities enhances the model's ability to capture complex interactions within the network, leading to more accurate and effective intrusion detection. Moreover, the framework not only provides performant detection capabilities but also delivers detailed, human-readable explanations and potential remedial actions, ensuring that the insights generated are both actionable and understandable. The framework is composed of six key components, as illustrated in Fig. 1.

The system initiates by capturing raw network traffic data (network packets), which are then processed by the first component, the Flow and Feature Generator. This component is responsible for generating network flows and extracting relevant features from both the flow and packet levels. These generated flows are subsequently passed to the Explainable Feature Extractor, which derives new features based on the temporal information from previous flows.

Once the features have been generated and enhanced, the data is transferred to the Graph Generator. This component transforms the flow and packet-level features into a heterogeneous graph structure, which serves as the foundation for the subsequent inference process. The transformed graph is then analyzed by the GNN Model, the fourth component, which performs the core task of intrusion detection by classifying the network traffic as either benign or respective attack class.

The output of the GNN Model is forwarded to both the Integrated Gradient Explainer and the Generative Explainer, components five and six, respectively. The Integrated Gradient Explainer identifies the most significant attributes of the graph structure that contributed to the model's prediction, providing a localized explanation of the model's output.

Leveraging these local explanations, the Generative Explainer generates specific prompts and formulates detailed, human-readable explanations of the results. Also, it offers potential remedial actions, ensuring that the system's outputs are not only accurate but also actionable and understandable.

The following sub-sections provide the technical details of each component, offering a comprehensive understanding of how our framework is utilized for effective cyber threat detection and explanation.

3.1.1. Flow and Feature Generator

The Flow and Feature Generator is the first component of our proposed framework, responsible for processing raw network traffic and aggregating it into flows. The primary goal of this component is to extract features from raw network traffic for real-time inference. For our approach, we set a maximum limit of 20 packets per flow, following the work of the CIC-IoT2023 dataset (Neto et al., 2023). The decision to limit the flow to 20 packets is driven by our objective to enable real-time inference; allowing flows to accumulate based on default parameters could result in flow

Table 1
Explainable Feature Set for Time-Based Network Intrusion Detection

Feature Name	Description
Rolling_UDP_Sum	Cumulative count of UDP packets received at the destination within a rolling time window.
Rolling_TCP_Sum	Cumulative count of TCP packets received at the destination within a rolling time window.
Rolling_ACK_Sum	Cumulative count of TCP acknowledgement packets received at the destination within a rolling time window.
Rolling_FIN_Sum	Cumulative count of TCP FIN packets received at the destination within a rolling time window.
Rolling_RST_Sum	Cumulative count of TCP RST packets received at the destination within a rolling time window.
Rolling_fin_Sum	Cumulative count of TCP FIN packets received at the destination within a rolling time window.
Rolling_psh_Sum	Cumulative count of TCP PSH packets received by the destination within a rolling time window.
Rolling_SYN_Sum	Cumulative count of TCP SYN packets received by the destination within a rolling time window.
Rolling_ICMP_Sum	Cumulative count of ICMP requests received by the destination within a rolling time window.
Rolling_http_port	Frequency of access attempts to well-known HTTP ports at the destination within a rolling time window.
Rolling_Average_Duration	Average duration of bidirectional communication sessions with the destination within a rolling time window.
Rolling_DNS_Sum	Cumulative count of DNS requests received by the destination within a rolling time window.
Rolling_vulnerable_port	Indicates the presence of known vulnerable ports at the destination within a rolling time window.
Rolling_packets_Sum	Cumulative count of all packets received at the destination within a rolling time window.
Rolling_bipackets_Sum	Cumulative count of bidirectional packets (both incoming and outgoing) associated with the destination within a rolling time window.
Unique_Ports_In_SourceDestination	Tracks the number of unique source ports used to communicate with a specific destination within a rolling time window.

durations of up to 30 minutes (Aouini and Pekar, 2022). Additionally, we set an idle timeout of 120 seconds, meaning that if a flow remains inactive for this duration, it is concluded.

The Flow and Feature Generator also **segregates** packet-level information with respect to its corresponding flow, **ensuring that each flow includes payload information derived from its associated packets**. This segregation is crucial for generating graph structures, as our objective is to unify flow and packet-level information into a cohesive representation.

Built upon NFStream (Aouini and Pekar, 2022), the **Flow and Feature Generator computes 76 flow-level features and 14 packet-level features**. The primary focus of the packet features is to capture the payload information, which is crucial for detecting payload-specific attacks.

Once the flows and their respective features are generated, the Flow and Feature Generator transforms the payload of each packet within a flow into a uniform feature space of 1500 features. This transformation is based on the method outlined in Farrukh et al. (2022), where **the payload of each packet is represented by 1500 features derived from the bytes of the payload**. The **hexadecimal** representation of **each byte is converted into an integer ranging from 0 to 255**, with each resulting integer forming one feature. In cases where the payload is shorter than 1500 bytes, **zero padding** is applied to maintain a consistent feature vector structure. For packets with no payload, the entire feature vector is padded with zeros.

Mathematically, the feature space for **a flow** can be represented as follows:

$$\text{Features} = [F_1, F_2, \dots, F_{76}, P_{\text{payload}}, P_{\text{flag}}, \dots, P_{\text{layersize}}] \quad (1)$$

where:

$$P_{\text{payload}} = [P_1^{1500}, P_2^{1500}, \dots, P_n^{1500}]$$

$$P_{\text{flag}} = [P_1^x, P_2^x, \dots, P_n^x]$$

$$P_{\text{layersize}} = [P_1^x, P_2^x, \dots, P_n^x]$$

In this context:

- $F_1, F_2, \dots, F_{76}$ denote the 76 flow-level features, encapsulating various attributes related to the overall network flow.
- $P_{\text{payload}} = [P_1^{1500}, P_2^{1500}, \dots, P_n^{1500}]$ represents the payload features of the packets within the flow, where each packet P_i is expressed as a 1500-dimensional vector. The superscript 1500 denotes the dimensionality of each payload vector.
- $P_{\text{flag}} = [P_1^x, P_2^x, \dots, P_n^x]$ represents another set of packet-level features, such as flags, with dimensionality x for each packet.
- $P_{\text{layersize}} = [P_1^x, P_2^x, \dots, P_n^x]$ represents additional packet-level features, such as IP layer size, also with dimensionality x for each packet.

Algorithm 1 Explainable Feature Extractor for NIDS

Input:

- 1: $W \rightarrow$ Rolling time window
- 2: $F \rightarrow$ Conventional flow features

Output:

- 3: $E \rightarrow$ Extracted features for NIDS

Step 1: Initialize Sliding Window Tracking

- 4: **for** each destination D_j **do**
- 5: Initialize $W_j \leftarrow [0, 0, \dots, 0]$ (size W)
- 6: **end for**

Step 2: Update Sliding Window and Calculate Temporal Features

- 7: **for** each time step t_i **do**
- 8: **for** each destination D_j **do**
- 9: Update W_j with packets received at t_i
- 10: Calculate temporal features T_j from W_j
- 11: **end for**
- 12: **end for**

Step 3: Integrate Conventional and Temporal Features

- 13: $E \leftarrow$ Combine F and T for all destinations

Return:

- 14: Extracted features E

3.1.2. Explainable Feature Extractor

Conventional features, developed by the first component, are typically effective for identifying volume-based attacks characterized by significant spikes in SYN requests or packet rates within a single flow but are less effective in detecting attacks that require understanding patterns across multiple flows. To address this limitation, the explainable feature extractor also extracts temporal features that capture statistics from previous flows. This approach involves analyzing the evolution of network flow statistics over time and identifying patterns across multiple flows. Techniques such as sliding window features, which aggregate statistics within specific time frames (e.g., connection attempts or packet rates over the last minute), provide a more comprehensive view of network activity and help identify deviations from normal behavior. Temporal correlation, which tracks the relationship between successive flows such as repeated scanning attempts from the same source, aids in identifying reconnaissance activities. By integrating these additional features, the detection system can better understand and explain the broader context of network traffic patterns, leading to a more performant approach for identifying and mitigating threats.

Our proposed approach for extracting additional features is illustrated in Algorithm 1. This approach leverages sliding window and temporal techniques to enhance the detection and explanation of cyber-attacks. As detailed in Table 1, the algorithm tracks and calculates various rolling window-based temporal features for each destination machine. These temporal features are systematically updated over time and are then integrated with conventional flow features. The resulting set of metrics offers a comprehensive framework for network intrusion detection, improving both the accuracy and interpretability of the system's predictions. This integration of temporal and flow-based features allows for a more dynamic and context-aware understanding of network behavior, which is crucial for identifying and explaining complex, time-based attacks.

3.1.3. Graph Generator

To effectively utilize network traffic data for network intrusion detection, we transformed the extracted flow and packet features into a heterogeneous graph structure. This transformation leverages both flow-level and packet-level information, facilitating a comprehensive analysis of the network traffic. Our graph structure includes two types of nodes and two types of edges, enabling detailed and nuanced modeling of network activities.

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ represent our heterogeneous graph, where $\mathcal{V}$ is the set of nodes and $\mathcal{E}$ is the set of edges. The nodes in our graph are categorized into two types: flow nodes and packet nodes. Each flow node $v_f \in \mathcal{V}_f$ corresponds to a unique network flow $\mathcal{F}$, and each packet node $v_p \in \mathcal{V}_p$ represents an individual packet $\mathcal{P}_i$ within a flow $\mathcal{F}$ having n

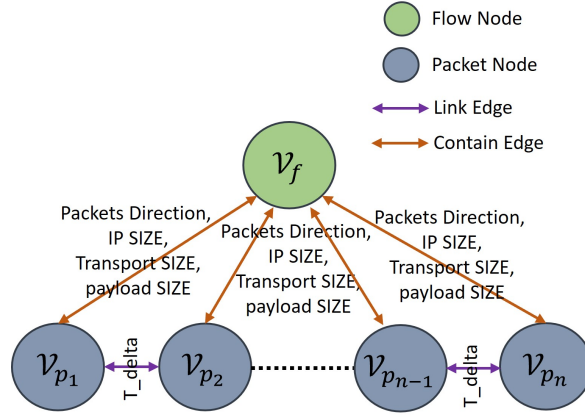


Fig. 2. Illustration of graph structure, fusing flow and packet level data modalities. The features of edges utilized in the generation of graph structures are highlighted on each edge. For 'link' edge we are utilizing only one feature whereas for the contain edge the number of features are four.

packets. Thus, the set of nodes $\mathcal{V}$ is given by:

$$\mathcal{V} = \mathcal{V}_f \cup \mathcal{V}_p \quad (2)$$

Each flow node v_f is associated with a feature vector $\mathbf{h}_{v_f} \in \mathbb{R}^d$, derived from flow-level attributes. Similarly, each packet node v_p has a feature vector $\mathbf{h}_{v_p} \in \mathbb{R}^{1500}$, where the attributes are primarily based on packet payload data.

Edges in the graph $\mathcal{E}$ also fall into two categories: "contain" edges $\mathcal{E}_c$ and "link" edges $\mathcal{E}_l$. A "contain" edge $e_c \in \mathcal{E}_c$ connects a flow node v_f to its corresponding packet nodes v_p . The feature vector for a "contain" edge, $\mathbf{a}_{e_c} \in \mathbb{R}^b$, includes the layer sizes and direction of the packet. The $\mathcal{E}_c$ are represented as:

$$\mathcal{E}_c = \{(v_f, v_{p_i}) \mid 1 \leq i \leq n\} \quad (3)$$

A "link" edge $e_l \in \mathcal{E}_l$ connects sequential packet nodes within a flow, thereby forming a directed acyclic graph (DAG). The feature vector for a "link" edge, $\mathbf{a}_{e_l}$, includes the time difference (t_δ) between two consecutive packets. $\mathcal{E}_l$ are represented as:

$$\mathcal{E}_l = \{(v_{p_i}, v_{p_{i+1}}) \mid 1 \leq i < n\} \quad (4)$$

The process of generating the graph structure is concisely presented in Algorithm 2. Additionally, we developed an open-source tool, GNN4ID (Farrukh et al., 2024b), which facilitates the transformation of raw network traffic into the proposed heterogeneous graph structure. GNN4ID effectively integrates the functionalities of our first three components, enabling users to seamlessly convert any raw network traffic data into the desired graph format. This tool comprehensively demonstrates the entire process, from extracting information from raw packet capture files to generating flow and packet-level features, and ultimately constructing the specified graph structure. A visual depiction of our graph structure is provided in Fig. 2, highlighting the detailed connections and attributes of both nodes and edges.

3.1.4. Graph Neural Network Model

The proposed HGNN model is designed to effectively process the dual modalities of network traffic packet-level and flow-level information—by leveraging a heterogeneous graph structure. The model is built upon the Graph Attention Convolution (GATConv) (Velićković et al., 2017) approach to capture the intricate relationships between different types of nodes and edges in the network traffic graph.

The layered architecture of the proposed HGNN model is as follows:

$$h_{v_i}^{(1)} = \text{ReLU} \left(\text{GATConv} \left(h_{v_i}^{(0)}, \mathbf{A}, \mathbf{E} \right) \right) \quad (5)$$

Algorithm 2 Generation of Network Traffic Graphs

Input: $\mathcal{F} \rightarrow \mathbf{h}_{v_f}, \mathbf{h}_{v_p}$ for $\mathcal{P}_i$ in $\mathcal{F}$
 1: **Initialize:** $\mathcal{V}$ and $\mathcal{E}$ as empty sets.
 2: **for** each packet $\mathcal{P}_i$ in flow $\mathcal{F}$ **do**
 3: Create a packet node v_p for $\mathcal{P}_i$.
 4: Add v_p to $\mathcal{V}_p$.
 5: **end for**
 6: Create a flow node v_f representing flow F .
 7: Add v_f to $\mathcal{V}_f$.
 8: **for** each packet node v_p corresponding to $\mathcal{P}_i$ **do**
 9: Add a "contain" edge e_c connecting v_f to v_p in $\mathcal{E}_c$.
 10: **end for**
 11: **for** each i from 1 to $|\mathcal{P}| - 1$ **do**
 12: Add a "link" edge e_l connecting v_{p_i} and $v_{p_{i+1}}$ in $\mathcal{E}_l$.
 13: **end for**
Return: The graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$.

where $h_{v_i}^{(0)}$ denotes the initial node features for each node type $\mathcal{V}_i$, $\mathbf{A}$ represents the adjacency matrix, and $\mathbf{E}$ denotes the edge features. The GATConv layers learn attention scores for each edge, enabling the model to focus on the most relevant connections in the graph.

In the HGNN model, both node and edge features are crucial for accurately capturing the characteristics of network traffic. The GATConv layers compute attention coefficients α_{ij} for each edge (i, j) , which are then used to aggregate the features of neighboring nodes and edges. The aggregation function for the node embeddings at layer l is defined as:

$$h_{v_i}^{(l)} = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(l)} \mathbf{W}^{(l)} h_{v_j}^{(l-1)} + \sum_{e_{ij} \in \mathcal{E}} \alpha_{ij}^{(l)} \mathbf{W}_e^{(l)} h_{e_{ij}}^{(l-1)} \right) \quad (6)$$

where $\mathcal{N}(i)$ represents the neighbors of node i , $\mathbf{W}^{(l)}$ and $\mathbf{W}_e^{(l)}$ are the learnable weight matrices for nodes and edges, respectively, and σ is a non-linear activation function (ReLU or LeakyReLU). The edge features $h_{e_{ij}}^{(l-1)}$ are combined with the node features from the previous layer to provide a richer representation at each layer. This combination allows the model to capture both local (node-level) and global (edge-level) interactions within the graph.

The HGNN model consists of two GATConv layers, each followed by a batch normalization step and a LeakyReLU activation function to introduce non-linearity:

$$h_{v_i}^{(2)} = \text{ReLU} \left(\text{BN} \left(\text{GATConv} \left(h_{v_i}^{(1)}, \mathbf{A}, \mathbf{E} \right) \right) \right) \quad (7)$$

Here, the batch normalization function $\text{BN}(\cdot)$ is applied to the output of the GATConv layer to stabilize the learning process, particularly in deep networks, by normalizing the output features. The LeakyReLU activation function ensures that the model can learn from both positive and negative feature values, thereby enhancing its ability to capture complex patterns in the data.

Following the two convolutional layers, the node embeddings are aggregated into a single graph-level embedding using a global mean pooling operation:

$$\mathbf{h}_{\text{graph}} = \text{GlobalMeanPool} \left(h_{v_i}^{(2)} \right) \quad (8)$$

This pooling step generates a unified representation of the entire graph by averaging the embeddings of all nodes, ensuring that both packet-level and flow-level information are comprehensively captured.

The graph-level embedding is then passed through a series of fully connected layers to produce the final classification output. These layers progressively reduce the dimensionality of the embedding while refining the learned features:

$$\text{Out} = \text{LogSoftmax} \left(\mathbf{W}_2 \cdot \text{ReLU} \left(\mathbf{W}_1 \cdot \text{ReLU} \left(\mathbf{W}_0 \cdot \mathbf{h}_{\text{graph}} \right) \right) \right) \quad (9)$$

where $\mathbf{W}_0$, $\mathbf{W}_1$, and $\mathbf{W}_2$ are weight matrices for the fully connected layers, and $\text{LogSoftmax}(\cdot)$ converts the final output into class probabilities.

The proposed HGNN model effectively integrates packet-level and flow-level information within a heterogeneous graph framework, combining node and edge features to capture the full spectrum of interactions in network traffic data. By utilizing GATConv layers with attention mechanisms, the model prioritizes critical connections within the graph, enhancing the overall accuracy of the classification.

3.1.5. Integrated Gradient Explainer

The fourth component in our system is the Integrated Gradient Explainer, which provides feature-based local explanations for each predicted outcome from the HGNN model. This explainer uses the Integrated Gradient approach to determine the contribution of each input feature to the network's prediction (Sundararajan et al., 2017). One of the primary advantages of Integrated Gradients is that it does not require any modification to the original network architecture. It is implemented with a few calls to the standard gradient operator, making it simple and efficient to use. Moreover, this method ensures that the attributions are accurate and meaningful, as it satisfies key theoretical principles that make the explanations reliable.

Integrated Gradients work by considering the path integral of the gradients of the prediction function F along a straight line from a baseline input to the actual input. If a baseline input is not provided, zero is used as the default value. The method computes the integral of the gradients at all points along this path, resulting in attributions that explain the importance of each feature in the input. Mathematically, this is represented as follows:

$$\text{IntegratedGrad}_i(x) = (x_i - x'_i) \times \int_{\alpha=0}^1 \frac{\partial F(x' + \alpha(x - x'))}{\partial x_i} d\alpha \quad (10)$$

Here, $\frac{\partial F}{\partial x_i}$ represents the gradient of the prediction function F with respect to the i -th input feature. x is the actual input, and x' is the baseline input. The integral accumulates these gradients along the path from x' to x , weighting them by the difference between x and x' along each feature dimension.

Function F represents the prediction function of the HGNN. It maps the input features of the heterogeneous graph to the output prediction. In our context, $F(x)$ is the predicted outcome based on the input features x , which include node attributes, edge weights, and other relevant information from the heterogeneous graph.

In our HGNN, the Integrated Gradient Explainer generates explanations by calculating the contributions of each feature in the heterogeneous graph to the network's prediction. This approach helps us understand which features are most influential for each specific prediction, thereby making the model's decisions more interpretable.

3.1.6. Generative Explainer

The generative explainer module uses a structured approach, integrating both flow and payload importance to create human-readable explanations. This process begins by utilizing the output of the Integrated Gradients Explainer to assess the importance of heterogeneous graph features. The extracted importance values are then used to create prompts, which are sent to the LLM (a Llama 3-8B model) to generate comprehensive explanations through zero-shot prompting.

Developing a structured prompt is particularly critical in zero-shot prompting approach because it allows the LLM to generate accurate and contextually relevant explanations without requiring additional training on specific examples. The effectiveness of zero-shot prompting relies heavily on the quality of the prompts. Therefore, developing well-structured prompts is essential to guide the LLM in producing meaningful outputs. The process of crafting these prompts is detailed in Algorithm 3.

The first step of this process involves initializing the prompt with a phrase that clearly states the predicted class from the HGNN. Specifically, we used the following initialization phrase (referred to as P_{init}): "The predicted class from GNN is {PredictedClass}." This initial prompt sets the context for the LLM to focus on the specific prediction made by the GNN. Following this, the flow importance values are processed by sorting them in descending order and selecting the top features that contributed most to the prediction. These features and their corresponding actual values are then integrated into the second part of the prompt, which is constructed to provide specific details about the top contributing factors. This segment of the prompt is framed as: "The top features contributing to this prediction are:" followed by a list of features and their actual values (referred to as P_{part2}).

Algorithm 3 Generative Explanation for GNN Predictions

Input:

- 1: $\mathbf{P}_{\text{imp}} \rightarrow$ Importance values for payload features
- 2: $\mathbf{P}_{\text{val}} \rightarrow$ Actual values of payload features
- 3: $\mathbf{F}_{\text{imp}} \rightarrow$ Importance values for flow features
- 4: $\mathbf{F}_{\text{val}} \rightarrow$ Actual values of flow features
- 5: $C_{\text{pred}} \rightarrow$ Class predicted by the GNN
- 6: $\mathbf{N}_{\text{feat}} \rightarrow$ Descriptive names of features

Output:

- 7: $\mathbf{G}_{\text{exp}} \rightarrow$ Explanation generated by the LLM
- 8: $P_{\text{init}} \leftarrow$ "The predicted class from GNN is $\{ C_{\text{pred}} \}$ "
- 9: $\mathbf{F}_{\text{sorted}} \leftarrow$ Sort $\mathbf{F}_{\text{imp}}$ in descending order
- 10: $\mathbf{F}_{\text{top}} \leftarrow$ Extract top n features from $\mathbf{F}_{\text{sorted}}$
- 11: $P_{\text{part2}} \leftarrow$ "The top features contributing to this prediction are:"
- 12: **for** each i in $\mathbf{F}_{\text{top}}$ **do**
- 13: $P_{\text{part2}} \leftarrow P_{\text{part2}} + \{ \mathbf{N}_{\text{feat}}[i] \}$ with actual value $\{ \mathbf{F}_{\text{val}}[i] \}$
- 14: **end for**
- 15: $P_{\text{align}} \leftarrow$ Alignment section phrase
- 16: $Q_{\text{flow}} \leftarrow P_{\text{init}} + P_{\text{part2}} + P_{\text{align}}$
- 17: $R_{\text{flow}} \leftarrow$ Send Q_{flow} to the LLM
- 18: **if** C_{pred} corresponds to payload-specific attacks **then**
- 19: $\mathbf{P}_{\text{norm}} \leftarrow$ Normalize $\mathbf{P}_{\text{imp}}$ vectors
- 20: $P_{\text{avg}} \leftarrow$ Calculate average importance of $\mathbf{P}_{\text{norm}}$
- 21: $\mathbf{P}_{\text{sorted}} \leftarrow$ Sort in $\mathbf{P}_{\text{avg}} \leftarrow$ descending order
- 22: $\mathbf{P}_{\text{top}} \leftarrow$ Extract top n payloads from $\mathbf{P}_{\text{sorted}}$
- 23: $P_{\text{hex}} \leftarrow$ Convert $\mathbf{P}_{\text{top}}$ from decimal to hex
- 24: $P_{\text{ascii}} \leftarrow$ Convert $\mathbf{P}_{\text{hex}}$ into ASCII string
- 25: $P_{\text{payloadPrefix}} \rightarrow$ Payload analysis phrase
- 26: $Q_{\text{payload}} \leftarrow P_{\text{payloadPrefix}} + P_{\text{ascii}} + P_{\text{align}}$
- 27: $R_{\text{payload}} \leftarrow$ Send Q_{payload} to the LLM
- 28: **end if**
- 29: **Return** $\mathbf{G}_{\text{exp}} \leftarrow R_{\text{flow}} + R_{\text{payload}}$

An alignment section is then added to the prompt. The alignment phrase (denoted as P_{align}) is crucial as it instructs the LLM to focus on explaining the predicted outcome and its potential reasons without introducing new or unrelated information. The phrase used is: "Don't expect any values on your own. Explain the predicted outcome and its potential reason along with the potential mitigation. Start your answer with "The predicted outcome is."

These parts—(P_{init} , P_{part2} , and P_{align})—are then combined to form a comprehensive query (Q_{flow}), which is then sent to the LLM to generate the flow-based response. If the GNN predicts a payload-specific attack, such as web-based or brute-force attacks, additional processing is done for the payload data. In this processing, the payload importance vectors are normalized, and the top payloads are converted into a human-readable ASCII string. Consequently, a second query (Q_{payload}) is constructed to analyze the payload data. This query is prefixed with the phrase assigned to $P_{\text{payloadPrefix}}$: "Analyze whether this payload of network flow is malicious or not. Give reason concisely." This query, along with the alignment section (P_{align}), is sent to the LLM to generate a corresponding response.

Finally, the outputs from the flow and payload-based queries are combined to form the complete generative explanation ($\mathbf{G}_{\text{exp}}$). This approach ensures that the explanations are accurate and understandable for both flow-based and payload-specific attacks.

3.2. Dataset

The CIC-IoT2023 dataset (Neto et al., 2023), developed by the Canadian Institute for Cybersecurity (CIC), is utilized to evaluate the effectiveness of our proposed framework. This dataset has been developed to capture the complexities and security challenges inherent in contemporary IoT networks, which are increasingly vulnerable to

Table 2

Description of the CICIoT2023 Dataset.

Type	Target	Total Number of Records	Class Distribution
Benign	Benign	1,098,195	2.35%
DDoS	Attack	33,984,560	72.79%
DoS	Attack	8,090,738	17.33%
Mirai	Attack	2,634,124	5.64%
Recon	Attack	354,565	0.76%
Spoofing	Attack	486,504	1.04%
WebBased	Attack	24,829	0.05%
Bruteforce	Attack	13,064	0.03%
Total		46,686,579	100%

Table 3

Attackers' Device Names and MAC Addresses

Device Name	MAC Address
Raspberry Pi 4—4 GB	E4:5F:01:55:90:C4
Raspberry Pi 4—2 GB	DC:A6:32:C9:E4:D5
Raspberry Pi 4—8 GB	DC:A6:32:DC:27:D5
Raspberry Pi 4—2 GB	DC:A6:32:C9:E5:EF
Raspberry Pi 4—2 GB	DC:A6:32:C9:E4:AB
Raspberry Pi 4—2 GB	DC:A6:32:C9:E4:90
Raspberry Pi 4—2 GB	DC:A6:32:C9:E5:A4
Ring Base Station	B0:09:DA:3E:82:6C
Fibaro Home Center Lite	AC:17:02:05:34:27

sophisticated cyber-attacks. This dataset is distinguished by its extensive coverage of network traffic, incorporating a wide array of IoT devices and diverse attack scenarios. Generated using a large-scale IoT topology comprising 105 devices, the dataset accurately reflects the interconnected and dynamic nature of modern IoT environments.

The dataset features 33 distinct attacks, systematically categorized into seven classes: Distributed Denial of Service (DDoS), Denial of Service (DoS), Reconnaissance, Web-based attacks, Brute Force attacks, Spoofing, and the Mirai botnet. These attacks were orchestrated by malicious IoT devices targeting other IoT devices, thereby creating realistic adversarial scenarios that are representative of actual threats faced by IoT ecosystems.

CIC-IoT2023 is provided in two formats: raw network traffic data (in pcap files) and an extracted flow-based dataset, which is computed within a fixed-size packet window. The dataset is vast, comprising a total of 46,686,579 events and featuring 47 distinct attributes. Table 2 offers a comprehensive breakdown of the CIC-IoT2023 dataset, detailing the types of attacks, the targets of these attacks, the total number of records, the distribution of records used for training and validation, and the percentage distribution across different classes. The "total number of records" specifically refers to the quantity of feature tuples extracted from the original pcap files, which are summarized within a fixed-size packet window.

3.2.1. Dataset Preprocessing

The data preprocessing stage involved two tasks to ensure the quality and balance of the dataset used for our analysis. The first task was focused on filtering the generated flows from GNN4ID based on the MAC addresses of the attackers provided in the CIC-IoT2023 dataset (Neto et al., 2023). We applied a filtering process where any flow instance with either the source or destination MAC address matching one of the known attacker addresses was retained and rest was removed from attack classes. Conversely, any flow associated with these attacker MAC addresses was excluded from the benign class. The specific MAC addresses identified as attackers are detailed in Table 3.

The second task involved addressing the significant class imbalance present in the CIC-IoT2023 dataset. Given the uneven distribution of classes, a combination of undersampling and oversampling techniques was employed to create a balanced dataset. Initially, 20% of the data samples were set aside to form the test set, with majority classes capped at 4,000 samples for this purpose. The remaining data were then adjusted, either through undersampling the majority

Table 4
Data Distribution After Data Preprocessing

Type	Flows After Filter	Test Set (20%)	Train Set (80%)
Benign	1,306,976	4,000	20,000
DDoS	33,137,785	4,000	20,000
DoS	7,889,155	4,000	20,000
Mirai	2,568,491	4,000	20,000
Recon	931,805	4,000	20,000
Spoofing	55,807	4,000	20,000
WebBased	5,449	1,090	20,000
Bruteforce	2,336	467	20,000

classes or oversampling the minority classes, to construct a training set of 20,000 samples per class. This approach ensured a more balanced distribution of samples across all classes, as illustrated in Table 4.

An important aspect of our preprocessing approach was the careful handling of attack subclasses. Since each main attack class in the CIC-IoT2023 dataset contains several subclasses, we ensured that the sampling process maintained proportional representation across these subclasses. This approach allowed us to preserve the diversity of attack types while achieving a more coherent and balanced dataset.

4. Results and Analysis

This section presents the results and analysis of our proposed framework, divided into two sub-headings: Performance Analysis and Explainability Analysis. The Performance Analysis subsection provides a comprehensive evaluation of the proposed approach, comparing its effectiveness against baseline models and state-of-the-art techniques. This comparison is aimed at demonstrating the superiority and robustness of our model in detecting network intrusions by fusing packet and flow level information together. Conversely, the Explainability Analysis subsection explores how our framework leverages contextual information to produce explainable and actionable insights, thereby enhancing the interpretability of the model's predictions and facilitating informed decision-making by security operators.

It is important to note that all results presented in this section are based on multi-class classification, encompassing eight distinct categories from the CIC-IoT2023 dataset. These evaluations were conducted under consistent experimental settings, ensuring identical data distribution and processing across all evaluation.

4.1. Performance Analysis

Conducting an apple-to-apple comparison for the proposed framework presents certain challenges, as our work is the first to fuse dual modalities—packet-level and flow-level information of network traffic—through a heterogeneous GNN, as per the authors knowledge. Consequently, comparing our framework with other approaches that solely utilize either flow-level or packet-level information may not provide an accurate assessment of its effectiveness. Furthermore, comparing our proposed HGNN with other graph-based approaches is also problematic, as the majority of existing works in NIDS utilizing GNNs focus on edge or node-level classification. These methods primarily aim at analyzing network traffic rather than providing real-time inference, as highlighted in Section 2. Therefore, comparing our graph-level classification with such node and packet-level classification makes no sense.

Given that our approach emphasizes real-time inference with a focus on effectively fusing dual modalities of network traffic to improve detection of both flow-level and packet-level attacks, a direct comparison with existing methods is inherently complex. Therefore, to provide a comprehensive evaluation of our approach, we divided our performance analysis into two distinct parts.

In the first part, we compare our framework with baseline models that either utilize flow information or payload information, offering a broad comparison with existing approaches. In the second part, we focus on state-of-the-art approaches that combine flow and packet-level information in some form, which allows us to evaluate our approach against more relevant benchmarks. This two-part analysis provides a balanced and thorough assessment of our framework's effectiveness.

4.1.1. Baseline Comparisons

In evaluating the baseline models, our objective is to offer a holistic view of the proposed framework when compared to approaches that solely rely on either flow-level or packet-level information. To provide deeper insights into the performance of the proposed framework, we also assessed the effectiveness of these baseline models on specific attack types—namely, payload-dependent attacks (such as brute force and web-based attacks) when relying on flow-level information, and payload-independent attacks when relying on packet-level information.

We conducted experiments using several widely adopted machine learning models, including Random Forest, Logistic Regression, AdaBoost Classifier, Multilayer Perceptron, K-Nearest Neighbors (KNN), and a simple three-layered Deep Neural Network (DNN) with default parameters. These evaluations were performed separately using flow-level information and packet-level information to benchmark these models. For utilizing packet-level information, we followed the approach outlined in Farrukh et al. (2022) to extract payload data and train the models accordingly.

Additionally, we incorporated several established approaches from the literature that use either flow-level or packet-level information, offering a more comprehensive comparison. Table 5 presents an overview of the comparison with approaches relying on flow-level information. The "Payload-Specific" column highlights the performance of the models on attacks that depend on the packet's payload. Similarly, Table 6 provides a comparison with approaches that utilize packet-level information, where the "Flow-Specific" column shows the performance on attack classes that are independent of payload data.

The results clearly demonstrate that our proposed framework outperforms the baseline models in both scenarios, underscoring the significance of incorporating dual modalities of network traffic. It is evident that approaches based solely on flow-level information underperform on attacks reliant on payload data, while approaches based on packet-level information exhibit weaknesses in detecting attacks that are not dependent on payload information. This highlights the necessity and effectiveness of our proposed dual-modality approach.

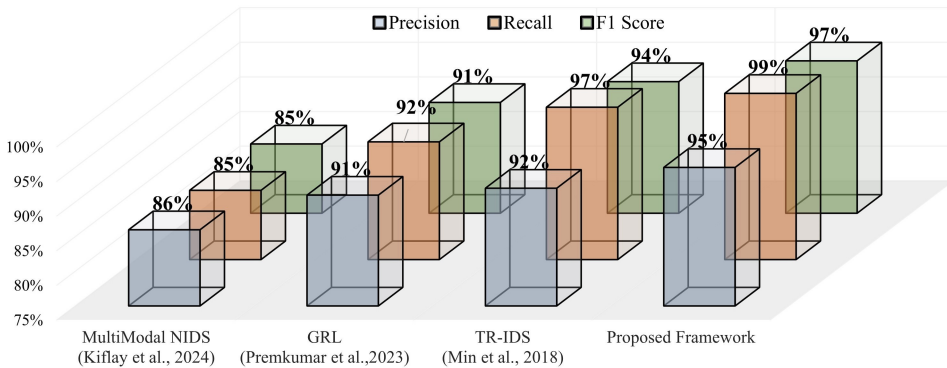


Fig. 3. Performance comparison between the proposed framework and several state-of-the-art models that incorporate dual modalities of network traffic. The proposed framework outperforms all other approaches, achieving a 97% F1-Score—a metric widely recognized for evaluating multi-class classification.

4.1.2. Dual-Modality State-of-the-Art

One of the standout aspects of our proposed framework is the fusion of dual modalities—packet-level and flow-level information—in network security. This multi-modal data fusion represents a significant innovation but also presents a challenge in evaluating our framework against other approaches, as the integration of both types of information is quite uncommon in the existing literature. Despite this, we identified a few state-of-the-art approaches that, in some capacity, utilize both packet-level and flow-level information, albeit through multiple steps or separate processes.

One such approach is presented by Premkumar et al. (2023), where the authors utilized packet-level information—specifically, the payload of packets—to compute embeddings of the packets within a flow. After generating these embeddings using a GNN, they combined the embeddings with respective flow features to perform classification, thereby leveraging dual modality. Another notable work by Kiflay et al. (2024) employs a two-step classification process. In their approach, two separate models are trained: one for flow-level information and another for packet-level information. This multimodal network traffic analysis approach, though effective, treats the modalities separately before combining them for final classification. Further, the TR-IDS framework proposed by Min et al. (2018) incorporates a

Table 5

Comparison with Approaches Using Flow-level Information.

Methods	Performance Metrics			Payload-Specific
	Precision	Recall	F1 Score	
Random Forest	0.97	0.96	0.96	0.92
Logistic Regression	0.82	0.91	0.85	0.78
Adaboost-Classifer	0.42	0.39	0.41	0.2
Multilayer Perceptron	0.87	0.93	0.9	0.81
KNeighborsClassifier	0.92	0.94	0.93	0.8
DNN	0.93	0.95	0.94	0.88
Conv-AE (Khan and Kim, 2020)	0.89	0.8	0.86	0.84
IIDS (Narayan et al., 2023)	0.77	0.92	0.81	0.69
CNN-LSTM (Gueriani et al., 2024)	0.95	0.96	0.96	0.86
Proposed Work	0.95	0.99	0.97	0.94

Table 6

Comparison with Approaches Using Packet-level Information.

Methods	Performance Metrics			Flow-Specific
	Precision	Recall	F1 Score	
Random Forest	0.82	0.64	0.74	0.71
Logistic Regression	0.64	0.52	0.53	0.55
Adaboost-Classifer	0.47	0.39	0.35	0.41
Multilayer Perceptron	0.78	0.57	0.62	0.64
KNeighborsClassifier	0.84	0.63	0.7	0.66
DNN	0.84	0.75	0.78	0.74
AEIDS (Pratomo et al., 2018)	0.79	0.64	0.72	0.64
Packet2Vec (Goodman et al., 2020)	0.68	0.6	0.62	0.62
EsPADA (Vidal et al., 2020)	0.64	0.6	0.62	0.58
Proposed Work	0.95	0.99	0.97	0.98

multi-stage processing pipeline. It uses word embeddings and a Text-CNN to extract features from packet payloads, which are then combined with flow features for classification. While these methods represent significant strides in multi-modal network security, they all rely on multi-step or multi-process approaches to fuse the dual modalities.

In contrast, our proposed framework inherently fuses packet-level and flow-level information within a unified, heterogeneous graph structure, eliminating the need for additional processing steps. This seamless integration not only simplifies the processing pipeline but also enhances the model's ability to detect sophisticated attacks that span both modalities. Moreover, our framework represents the first of its kind to fuse dual modalities through a heterogeneous graph, providing a novel and more holistic approach to network intrusion detection.

When compared with these state-of-the-art methods, our proposed framework demonstrates superior performance, as highlighted in Fig. 3. The inherent fusion of packet-level and flow-level information in our model allows for more accurate and efficient classification, setting a new benchmark in network intrusion detection. Our results clearly indicate that by integrating both modalities within a single framework, we can achieve better detection rates and more actionable insights than existing approaches, solidifying our framework's position as a cutting-edge solution in the field.

4.2. Explainability Analysis

To assess the explainability of our proposed system, we utilized the default hyperparameters of the Llama 3 model (AI@Meta, 2024) and applied Algorithm 3 to generate prompts that guide the LLM in producing the desired responses. This algorithm involves executing different queries tailored for flow-based and payload-specific attacks. Therefore, we present two sample responses generated by our proposed system, illustrating these two distinct cases.

4.2.1. Flow-based Attacks' Explanation

In the first scenario, the predicted attack is a flow-based attack, specifically a DDoS attack. The attacker attempts to overwhelm network resources in a distributed manner within a specific timeframe. The Generative Explainer leveraged the top features identified by the previous component and provided a comprehensive response, as illustrated in Fig. 4. The initial part of the response highlights the predicted attack using descriptive feature names. It clearly shows how

The predicted attack is a Distributed Denial of Service (DDoS) attack.

The Graph Neural Network (GNN) has identified the following features that contribute to this prediction:

1. Total Packets received at the destination within last few milliseconds: The GNN has observed a high value of 6997.000 packets received at the destination within the last few milliseconds, which is indicative of a DDoS attack.
2. Total ICMP Packets received at the destination within last few milliseconds: The GNN has observed a high value of 350.000 ICMP packets received at the destination within the last few milliseconds, which is also indicative of a DDoS attack.
3. src2dst_bytes: The GNN has observed a high value of 1200.000 bytes received from a specific source to the destination within the last few milliseconds, which is indicative of a DDoS attack.
4. ICMP Packets raised by Specific Source to destination within last few milliseconds: The GNN has observed a high value of 350.000 ICMP packets raised by a specific source to the destination within the last few milliseconds, which is also indicative of a DDoS attack.
5. bidirectional_bytes: The GNN has observed a high value of 1200.000 bytes in both directions between two nodes, which is indicative of a DDoS attack.

These features are contributing to the prediction of a DDoS attack because they indicate a large amount of traffic being sent to a single destination, which can overwhelm the targeted system and cause it to become unavailable.

To mitigate a DDoS attack, there are several steps that can be taken, including:

1. Rate limiting: Implementing rate limits on incoming traffic can help prevent a DDoS attack by limiting the amount of traffic that can be sent to the targeted system.
2. Content delivery networks (CDNs): Using a CDN can help distribute traffic across multiple servers, making it more difficult for an attacker to overwhelm the targeted system.
3. Firewalls: Implementing firewalls can help block incoming traffic that is not legitimate and prevent a DDoS attack.
4. Network segmentation: Segmenting a network can help isolate the targeted system and prevent an attacker from accessing it directly.
5. Incident response planning: Having an incident response plan in place can help quickly identify and respond to a DDoS attack, minimizing the impact on the targeted system.

Fig. 4. Illustration of the generative explanation for a flow-based DDoS. The response generated by the proposed system incorporates explainable features and a rolling window analysis, providing a more comprehensive and accurate reasoning for the detected attack.

an increase in ICMP packets within the specific timeframe suggests that the attacker is attempting to overwhelm the targeted system. The second part outlines the potential mitigation steps that can be taken to address such attacks.

To compare our generated explanations with previous work by Khediri et al. (2024), we replicated their methodology by utilizing Shapley values of the predicted outcomes and their instruction tuning template. The generated response, as shown in Fig. 5, demonstrates that relying solely on conventional flow attributes is insufficient to establish a clear relationship with the detected attack. This limitation arises because individual flow attributes, such as acceptable packet size ranges, can be misleading when considered in isolation. In contrast, DDoS attacks are more effectively explained by analyzing the average behavior of packets over a specific timeframe rather than focusing on single flow attributes. This comparison highlights that the inclusion of our explainable features and the rolling window concept provides more accurate reasoning for flow-based attacks, making the responses generated by our proposed system significantly more meaningful.

4.2.2. Payload-based Attacks' Explanation

The second type of attack involves packet-level threats, where the actual maliciousness resides in the packet payload. Although these attacks may exhibit benign behavior in terms of network flow, their malicious intent becomes apparent through payload analysis. Previous researchers (Khediri et al., 2024; Ziems et al., 2023; Ali and Kostakos, 2023) have explained such attacks using only flow attributes, which is insufficient because flow data alone doesn't reveal the inherent maliciousness in most payload attacks, such as SQL injection or malware distribution.

To address this limitation, we introduced a second case in our Algorithm 3, specifically designed for payload-specific attacks. For these types of attacks, our approach generates two queries: one focused on analyzing the payload text to assess potential exploitation, and another based on flow attributes. This dual-query method enables our system to provide a more comprehensive explanation, as shown in Fig. 6. The first part of the response clearly indicates that the packet payload is malicious, containing a potential SQL injection query. This payload analysis effectively conveys the malicious nature of the predicted attack. The subsequent sections then provide flow-based explanations and suggest potential mitigation steps, ensuring a thorough understanding of the threat.

The predicted attack from Graph Neural Network is Distributed Denial of Service (DDoS) attack.

Top features contributing in this prediction are:

Feature 1: 'bidirectional_max_ps' contributes with SHAP value 0.1013 (actual value: 60.0)
 Feature 2: 'src2dst_bytes' contributes with SHAP value 0.0845 (actual value: 1200.0)
 Feature 3: 'bidirectional_mean_ps' contributes with SHAP value 0.0824 (actual value: 60.0)
 Feature 4: 'src2dst_packets' contributes with SHAP value 0.0784 (actual value: 20.0)
 Feature 5: 'src2dst_max_ps' contributes with SHAP value 0.0759 (actual value: 60.0)

For this type of attack, the features that contribute the most to the prediction are related to the volume of traffic flowing through the network. Specifically, the 'src2dst_bytes' and 'src2dst_packets' features indicate the amount of data being transmitted between sources and destinations, which can be used to identify potential DDoS attacks.

To mitigate a DDoS attack, there are several strategies that can be employed, including:

1. Rate limiting: This involves setting limits on the amount of traffic that can be transmitted through the network in a given time period.
2. Traffic filtering: This involves filtering out traffic that is deemed to be malicious or unnecessary.
3. Content delivery networks (CDNs): CDNs can help distribute traffic and reduce the load on the network, making it more difficult for attackers to overwhelm the network.
4. Cloud-based security services: These services can provide additional security measures, such as intrusion detection and prevention, to help protect the network from DDoS attacks.
5. Network segmentation: This involves dividing the network into smaller, more manageable segments, which can help reduce the impact of a DDoS attack.

In summary, the predicted attack from Graph Neural Network is a Distributed Denial of Service (DDoS) attack, and the top features contributing to this prediction are related to the volume of traffic flowing through the network. To mitigate a DDoS attack, there are several strategies that can be employed, including rate limiting, traffic filtering, CDNs, cloud-based security services, and network segmentation.

Fig. 5. Illustration of the generative explanation for a flow-based DDoS attack, highlighting the insufficiency of relying solely on conventional flow attributes.

The payload is malicious. It attempts to exploit a SQL injection vulnerability by using a UNION query to extract sensitive data, including usernames, last names, and passwords from the 'dvwa.guestbook' table. This action could lead to unauthorized access and compromise of the system.

The predicted attack is a Web-based attack, which means that the attacker is trying to exploit vulnerabilities in web applications or web servers. The top features contributing to this prediction are related to the volume of traffic and the patterns of communication between the source and destination hosts.

1. bidirectional_bytes: The high value of this feature indicates that there is a large amount of traffic flowing between the source and destination hosts, which is a common characteristic of web-based attacks.
2. bidirectional_min_ps: The high value of this feature suggests that the source host is sending a large number of packets to the destination host within a short period of time, which could be an indication of a web-based attack.
3. bidirectional_mean_ps: Similar to the previous feature, the high value of this feature suggests that the source host is sending a large number of packets to the destination host, which could be an indication of a web-based attack.
4. src2dst_min_ps: The high value of this feature indicates that the source host is sending a large number of packets to the destination host within a short period of time, which could be an indication of a web-based attack.
5. ACK Packets raised by Specific Source to destination within last few milliseconds are: The high value of this feature suggests that the source host is raising a large number of ACK packets to the destination host within a short period of time, which could be an indication of a web-based attack.

To mitigate this type of attack, you can implement security measures such as:

- * Implementing web application firewalls (WAFs) to detect and block malicious traffic.
- * Conducting regular security audits and vulnerability assessments to identify and patch vulnerabilities in web applications and web servers.
- * Implementing intrusion detection and prevention systems (IDPS) to detect and block suspicious traffic.
- * Educating users about safe browsing practices and how to avoid falling victim to web-based attacks.

In summary, the predicted attack is a Web-based attack, and the top features contributing to this prediction are related to the volume of traffic and the patterns of communication between the source and destination hosts. To mitigate this type of attack, you can implement security measures such as WAFs, security audits, IDPS, and user education.

Fig. 6. Comprehensive explanation of a packet-level attack, demonstrating the malicious nature of the packet payload. The figure shows the dual-query approach, where the payload analysis identifies potential SQL injection, followed by flow-based explanations and suggested mitigation steps, ensuring a thorough understanding of the attack scenario.

5. Conclusion

This work presented "XG-NID," a novel framework that advances the state-of-the-art in ML-based NIDS by integrating flow-level and packet-level data into a heterogeneous graph structure. By leveraging a HGNN for graph-level classification, XG-NID provides real-time inference capabilities while capturing the complex relationships inherent in network traffic. The incorporation of LLMs adds a layer of explainability, enabling the generation of human-readable explanations and actionable insights, which are critical for effective cybersecurity defense mechanisms.

Furthermore, the introduction of new flow features based on temporal information enhances the contextual understanding and explainability of the model's predictions. Our comprehensive evaluation demonstrates that XG-NID significantly outperforms existing baseline and state-of-the-art methods, achieving an impressive F1 score of 97% in multi-class classification. This underscores the efficacy of our approach in addressing the challenges of real-time intrusion detection while providing a robust and adaptive defense mechanism. Ultimately, XG-NID sets a new benchmark in the field of NIDS by combining innovative data fusion techniques with enhanced interpretability and real-time operational capabilities, offering a promising direction for future research and practical implementations in cybersecurity.

Future work will focus on extending XG-NID to multiple datasets to ensure its robustness across diverse network environments. Additionally, we aim to integrate it into an automated system capable of making autonomous decisions and providing rapid security responses, paving the way for intelligent, self-sustaining cybersecurity solutions.

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CRedit authorship contribution statement

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